

Technical Design Document (TDD) & Interface Control (ICD)

Doc Ref: TDD-OPTUM-2026-04	Owning Dept: MLOps & Healthcare Integration	Architecture: LangGraph + Qdrant + FHIR	Status: ARB SIGNED OFF
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1. Production PostgreSQL State Table DDL

```
CREATE TYPE pa_status_enum AS ENUM ('INGESTED', 'EXTRACTING', 'EXPEDITED_SIGN_OFF', 'EXCEPTION_QUEUE', 'ADJUDICATED');
CREATE TYPE criterion_status_enum AS ENUM ('MET', 'NOT_MET', 'INDETERMINATE');

CREATE TABLE prior_auth_requests (
  request_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  patient_id_deid VARCHAR(64) NOT NULL,
  cpt_code VARCHAR(10) NOT NULL,
  employer_group_id VARCHAR(32) NOT NULL,
  policy_id VARCHAR(64) NOT NULL,
  current_status pa_status_enum NOT NULL DEFAULT 'INGESTED',
  aggregate_confidence NUMERIC(4,3) CHECK (aggregate_confidence BETWEEN 0.000 AND 1.000),
  hitl_assigned_nurse_id VARCHAR(64),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE pa_criteria_extractions (
  extraction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  request_id UUID REFERENCES prior_auth_requests(request_id) ON DELETE CASCADE,
  criterion_id VARCHAR(64) NOT NULL,
  status criterion_status_enum NOT NULL,
  confidence_score NUMERIC(4,3) NOT NULL,
  source_sentence TEXT NOT NULL,
  char_start INT NOT NULL, char_end INT NOT NULL,
  ragas_faithfulness_score NUMERIC(4,3) NOT NULL
);
```

2. Inbound CDS Hook & Outbound ASC X12N 278 EDI Contract

```
// Inbound CDS Hook: order-sign
{
  "hook": "order-sign",
  "context": { "patientId": "Patient/DEID-89412", "draftOrders": ["ServiceRequest/CPT-27447"] }
}

// Outbound ASC X12N 278 EDI Determination Transaction
ISA*00*          *00*          *ZZ*OPTUMCLINICAL  *ZZ*AVAILITY          *260831*0926*^*00501*000008941*0*P*:-
BHT*0007*13*AUTH20269012*20260831*0926~
UM*AR*I*2*21:B*E~ (Status: APPROVED / Injected into Cirrus/Facets Claims Engine)
```

3. Engineering Peer Review & Architecture Sign-Off

ROLE	ENGINEER NAME	REVIEW VERDICT	SIGN-OFF DATE
Staff ML / NLP Engineer	Alexandre Dubois, PhD	APPROVED (Pydantic locked)	2026-03-24
Integration Lead	David Washington	APPROVED (X12 278 Validated)	2026-03-25
Lead SRE / Platform	Elena Rostova	APPROVED (P95 Latency < 30s)	2026-03-25